Mittuniversitetet
MID SWEDEN UNIVERSITY

DT081A – Signal and Image Processing

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Begränsad delning

Lecture 5
Digital Filter Design

Begränsad delning

Linear Time-Invariant Causal System

Linear Time-Invariant Causal System

- Linear System

  Input $x = a \cdot x_1 + b \cdot x_2 \Rightarrow$ Output $y = a \cdot y_1 + b \cdot y_2$

- Time-Invariant System

  Delayed Input $x(n) = x_1(n - n_0) \Rightarrow$ Delayed Output $y(n) = y_1(n - n_0)$

- Causal system

  "Cause comes before effect"

Impulse response $h(n) = \begin{cases} h^+(n), & n \geq 0 \\ 0, & n < 0 \end{cases}$



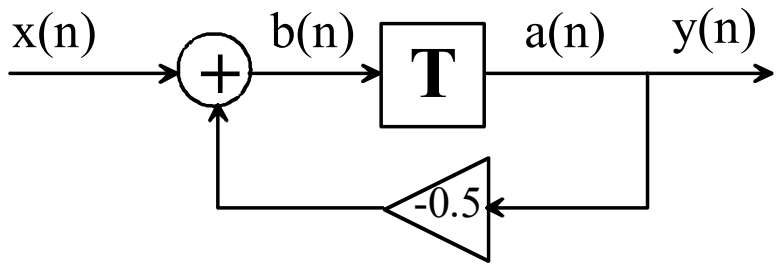
Output of an LTI System

- The output from a system is given by a convolution

$$y(n) = h(n) * x(n) = \sum_{k=-\infty}^{\infty} h(k) \cdot x(n-k) \qquad \text{(Discrete Time System)}$$

- An LTI system is fully defined by
  - Its *impulse response* $h(k)$
  - Or its transform $H(\cdot)$: Fourier Transform, Z-Transform, etc.
    - as they are invertible
  - Or its difference equation
    - Ex. $y(n) = x(n-1) - 0.5\, y(n-1)$





Output of an LTI System by an Impulse Input

- Let system input be an impulse (unit pulse)

$$x(n) = \delta_{\mathrm{K}}(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$ (impulse / unit pulse)

- The system output is then

$$y(n) = h(n) * \delta_{\mathrm{K}}(n) = \sum_{k=-\infty}^{\infty} h(k) \cdot \delta_{\mathrm{K}}(n-k) = h(k)$$ (unit pulse response a.k.a. impulse resp.)

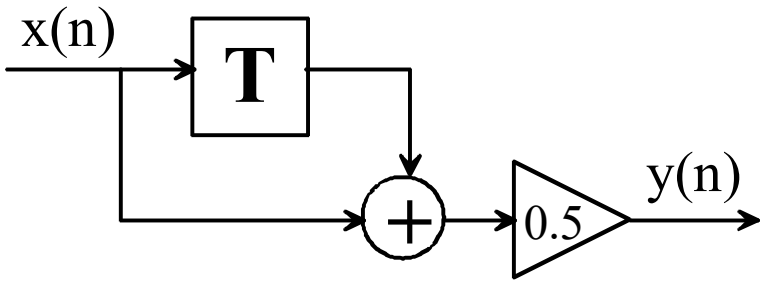
Begränsad delning

FIR System

Discrete Time System (Filter) – Example 1

- Average value
- What is the difference equation for this system?

$$y(n) = [x(n) + x(n-1)]/2$$



- What is the impulse response?

$$y(n) = \sum_{k=0}^{1} \frac{1}{2} x(n-k)$$

c.f. $y(n) = \sum_{k=-\infty}^{\infty} h(k) \cdot x(n-k)$

- Impulse response can be written

$$h(n) = \left[\frac{1}{2}\delta_K(n) + \frac{1}{2}\delta_K(n-1)\right]$$

- How long is the impulse response?
- This is a

Finite Impulse Response (FIR)

General FIR System (Filter)

- Difference equation for a FIR system

$$y(n) = \sum_{k=0}^{M} b_k \cdot x(n-k)$$

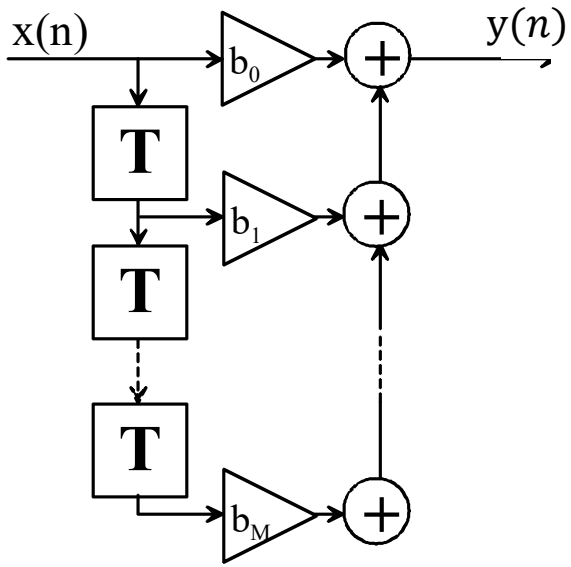
- What is the impulse response?

c.f. $y(n) = \sum_{k=-\infty}^{\infty} h(k) \cdot x(n-k)$

- Impulse response can be written

$$h(n) = b_n, \quad 0 \le n \le M$$

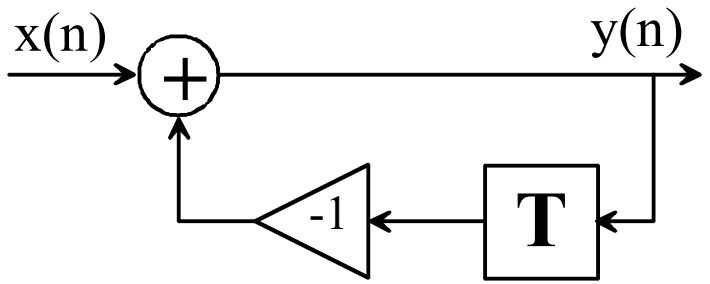
- Output depends on
  - Input $x(n)$



Begränsad delning

IIR System

Discrete Time System (Filter) – Example 2



- Feedback
- What is the difference equation for this system?

  $y(n) = x(n) - y(n-1)$

  Assume $y(n) = 0,\ n < 0$
- What is the impulse response?
  - Test with $x(n) = \delta_{\mathrm{K}}(n)$

$$y_{\delta_{\mathrm{K}}}(n) = \{1, -1,\ \ 1, -1,\ \ 1, -1,\ \cdots\}$$

$$\Longrightarrow h(n) = (-1)^n \cdot u(n)$$

- How long is the impulse response?
- This is an

  Infinite Impulse Response (IIR)

General Feedback System – IIR system (Filter)

- Difference equation for an IIR system

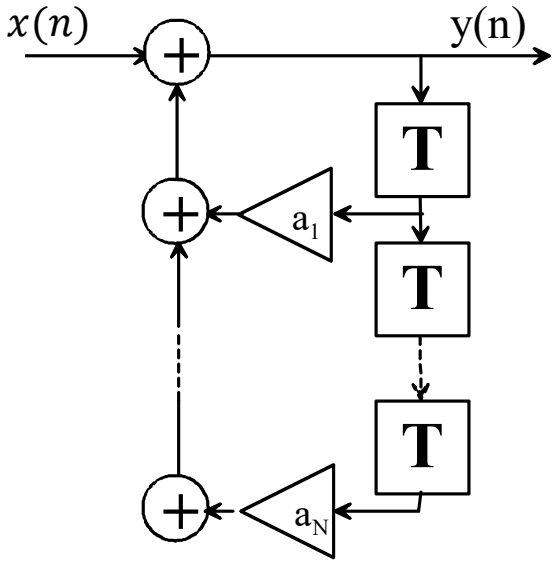
$$y(n) = \sum_{k=1}^{N} a_k \cdot y(n-k) \quad [+x(n)]$$

$a_k \neq 0$, for some k

- Impulse response can be written (infinite)

$$h(n) \neq 0, \quad C < n < \infty$$

- Output depends on
  - [Input $x(n)$]
  - Output $y(n-k)$, previous values

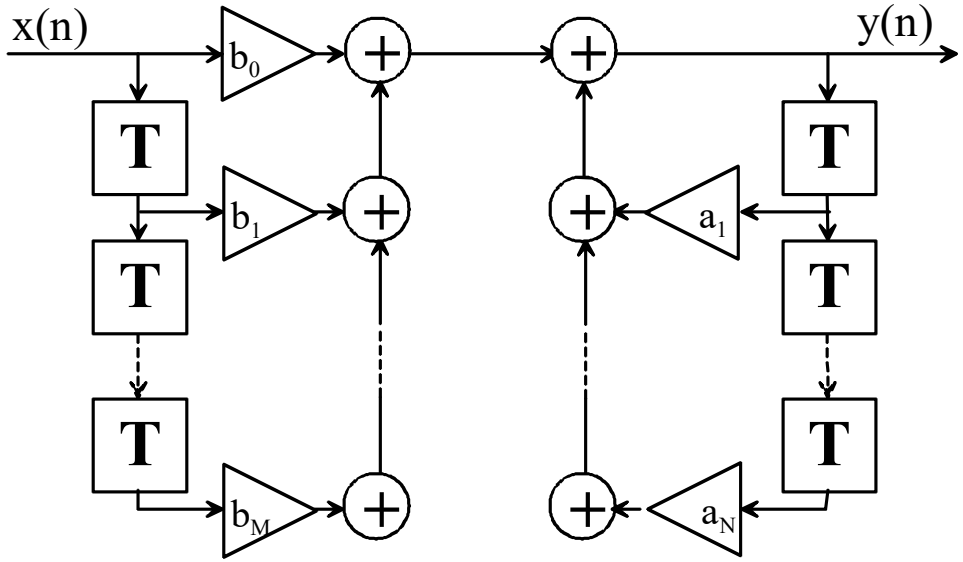


General IIR System (Filter)

- Difference equation for an IIR system

$$y(n) = \sum_{k=1}^{N} a_k \cdot y(n-k) + \sum_{k=0}^{M} b_k \cdot x(n-k)$$

$a_k \neq 0$, for some k



- Impulse response can be written (infinite)

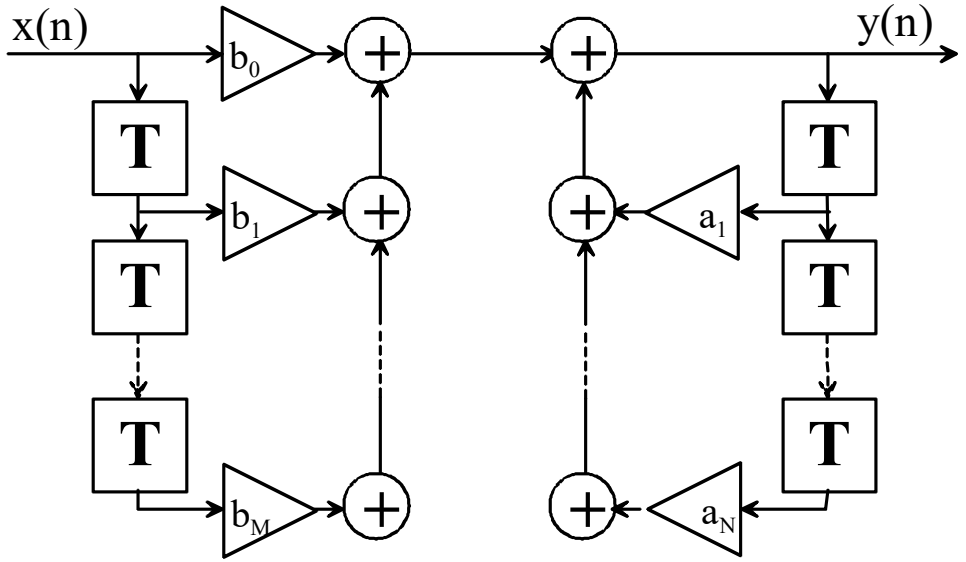
$$h(n) \neq 0, \quad C < n < \infty$$

- Output depends on
  - Input $x(n-k), k \geq 0$ for a causal system
  - Output $y(n-k), k \geq 1$, previous values

- By deciding $\{a_k\}$ and $\{b_k\}$ in an appropriate manner, a time-discrete system (filter) is obtained with desired properties

General IIR System (Filter) – Computational Complexity

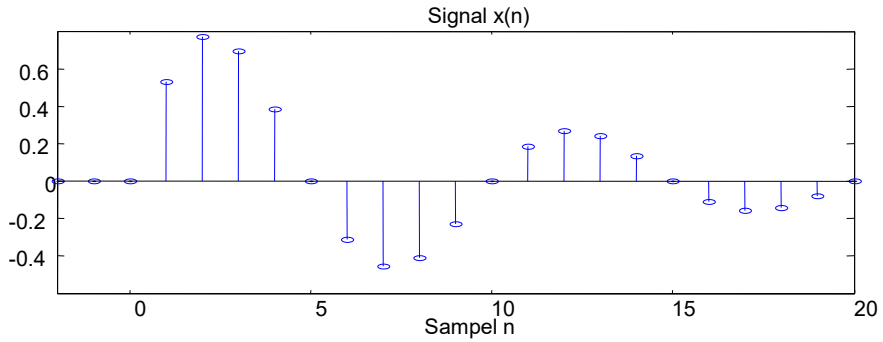
- How much memory is required?
- How many multiplications are required?
- How many additions are required?
- How many multiplications would an IIR filter require using a direct implementation of the impulse response?



Begränsad delning

Z-transform and Discrete-Time Fourier Transform

Z-transform



- The Z-transform is used for time-discrete sequences
  - $x(n) \in \mathbb{R},\ n \in \mathbb{Z}$
  - $x(n) = \{\dots, x(-1), x(0), x(1), x(2), \dots\}$

$$X(z) = Z\{x(n)\} = \sum_{n=-\infty}^{\infty} x(n) z^{-\mathrm{n}} \in \mathbb{C}$$

- Characteristics
  - Linearity

$$y(n) = a \cdot x_1(n) + b \cdot x_2(n) \ \leftrightarrow Y(z) = a \cdot X_1(z) + b \cdot X_2(z)$$

  - Time-shift

$$y(n) = x(n - m) \ \leftrightarrow Y(z) = z^{-m} X(z)$$

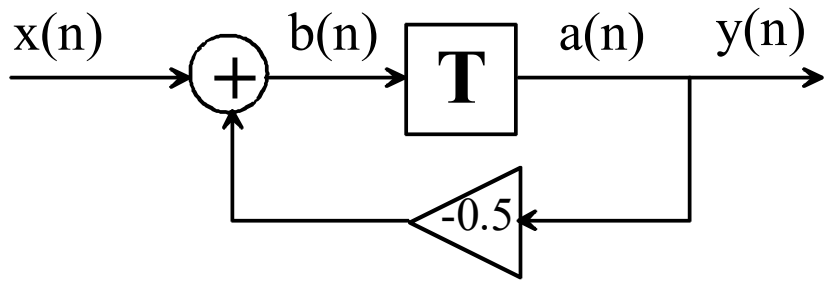
- Relation to discrete-time Fourier transform (requires convergence)

$$X\left(e^{j\Omega}\right) = X(z)\Big|_{z=e^{j\Omega}}$$

  - Thus we only consider the unit circle $|z| = 1$ of the complex z-space

Example – Discrete Time System

- Discrete Time System – Derivation



$$b(n) = a(n+1) = x(n) - 0.5\,y(n)$$
$$y(n) = a(n)$$

- Z-transform
  - $B(z) = z^1 A(z) = X(z) - 0.5\,Y(z)$
  - $Y(z) = A(z)$
- Solve for $Y(z)$

$$Y(z) = A(z) = z^{-1}X(z) - 0.5\,z^{-1}Y(z)$$
$$Y(z)(1 + 0.5\,z^{-1}) = z^{-1}X(z)$$

- Recall

$$y(n) = h(n) * x(n) \leftrightarrow Y(z) = H(z) \cdot X(z)$$

- The **System Transfer function** is

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z^{-1}}{1 + 0.5z^{-1}}$$

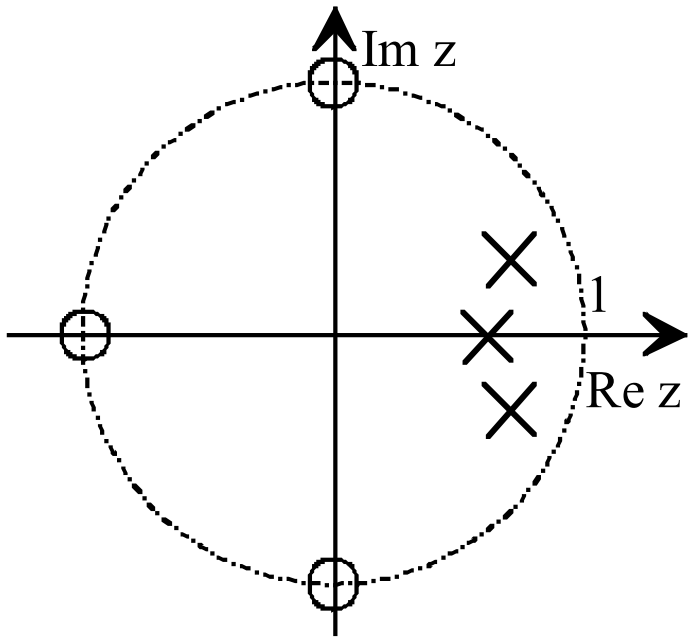
- The **System Frequency response** is
  - $H(e^{j\Omega}) = H(z)|_{z=e^{j\Omega}} = \frac{Y(z)}{X(z)}|_{z=e^{j\Omega}} = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})}$

Poles and Zeros of the System Transfer Function

- The System Transfer Function $H(z)$ can often be written as a rational expression of polynomials

$$H(z) = \frac{N(z)}{D(z)} = \frac{b_0 + b_1\, z^{-1} + b_2 z^{-2} + \cdots}{1 + a_1 z^{-1} + a_2 z^{-2} + \cdots}$$

- The solutions to $N(z) = 0$ are called *zeros*
  - Indicated by circles in the z-domain
- The solutions to $D(z) = 0$ are called *poles*
  - Indicated by crosses in the z-domain

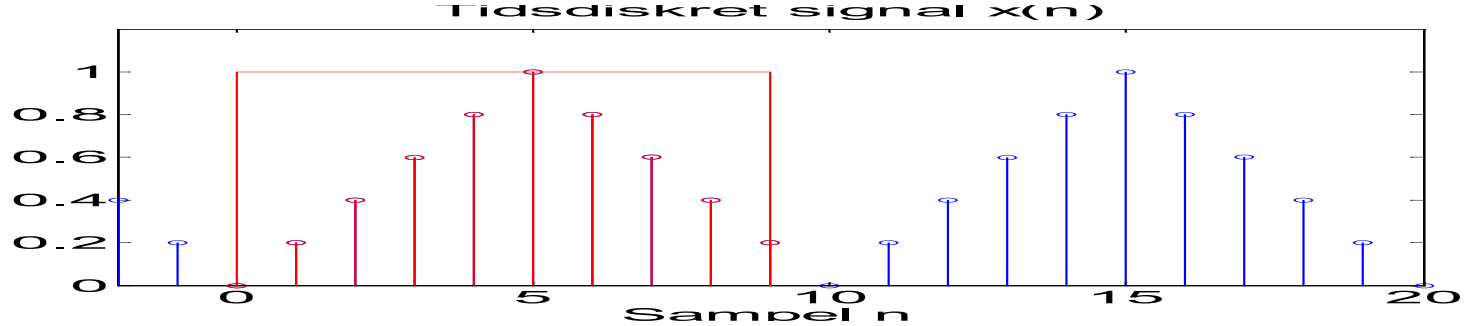


Begränsad delning

Windowing

Discrete-Time Fourier Transform

- Consider a causal signal $x(n) = \begin{cases} x^+(n), & n \geq 0 \\ 0, & n < 0 \end{cases}$
- DTFT : $X(e^{j\Omega}) = \sum_{n=0}^{\infty} x(n) e^{-j\Omega n} \in \mathbb{C}$
- In reality, only N samples can be measured and used.



- The truncated sequence corresponds to a multiplication with a window $w(n)$

$$x_{tr}(n) = x(n) \cdot w(n) \qquad w(n) = \begin{cases} 1, & 0 \leq n \leq N-1 \\ 0, & \text{else} \end{cases}$$

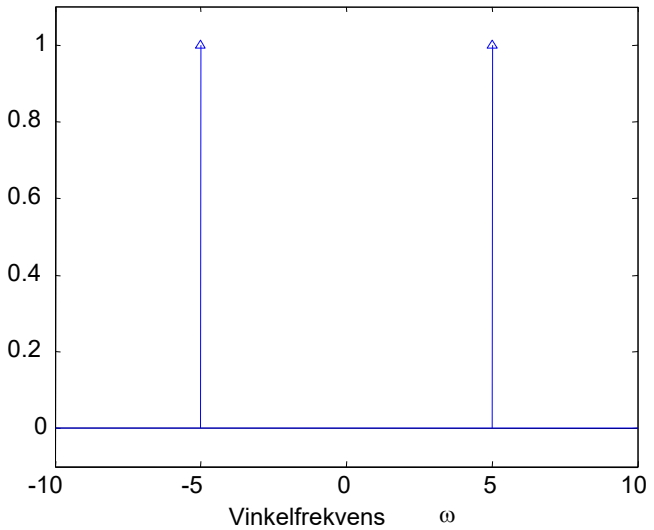
Windowing

- A multiplication in time-domain
- Corresponds to a convolution in frequency domain

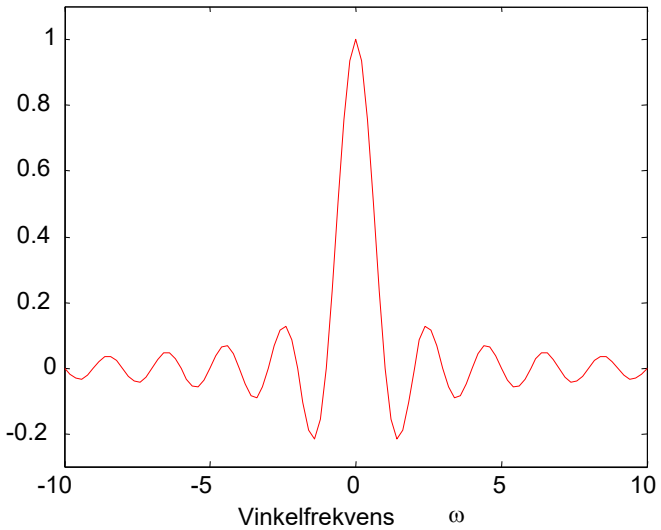
$$x_{tr}(n) = x(n) \cdot w(n)$$

$$X_{tr}(e^{j\Omega}) = X(e^{j\Omega}) * W(e^{j\Omega})$$

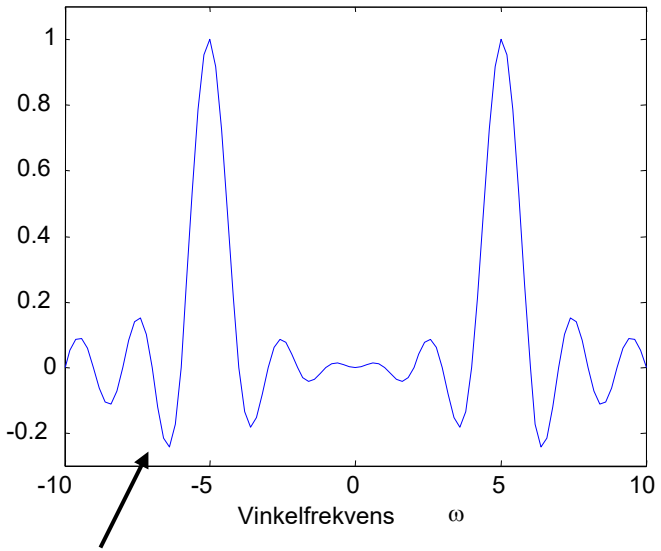
Amplitude spectrum $X(e^{j\Omega})$



Amplitude spectrum $W(e^{j\Omega})$



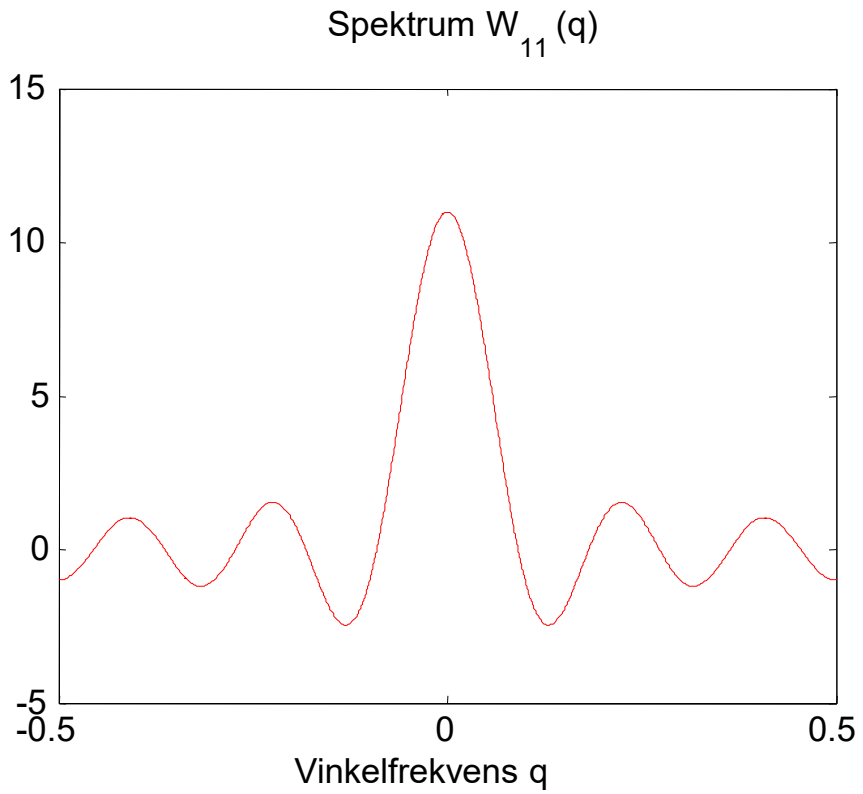
Amplitude spectrum $X_{tr}(e^{j\Omega})$

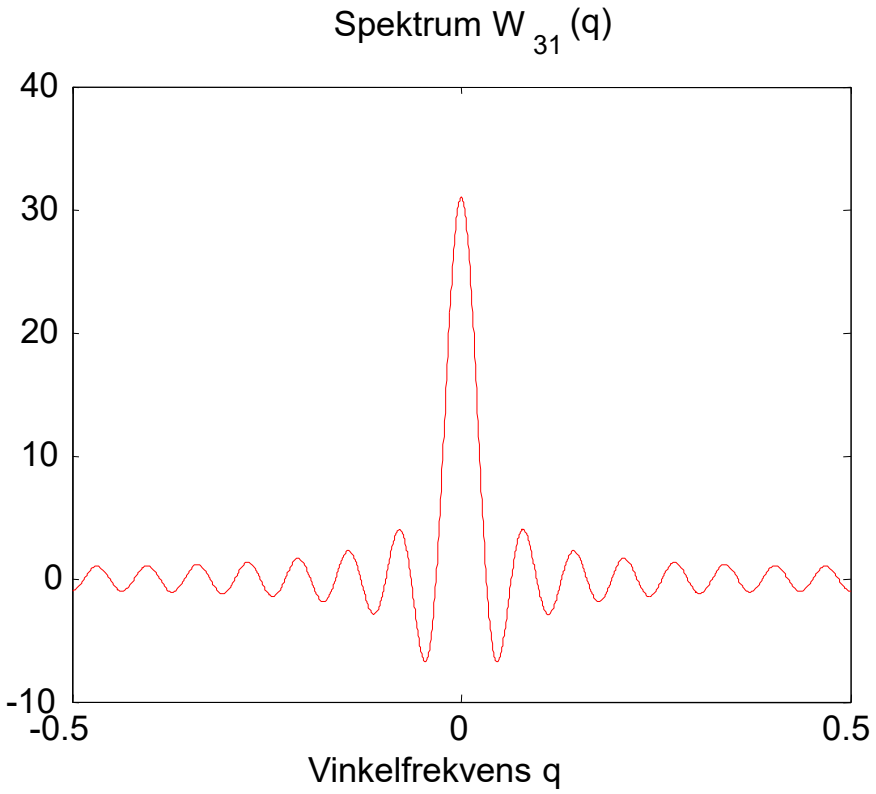


- *Leakage*: The signal $x_{tr}(n)$seems to have frequencies next to the actual frequencies

Windowing – Size of Window

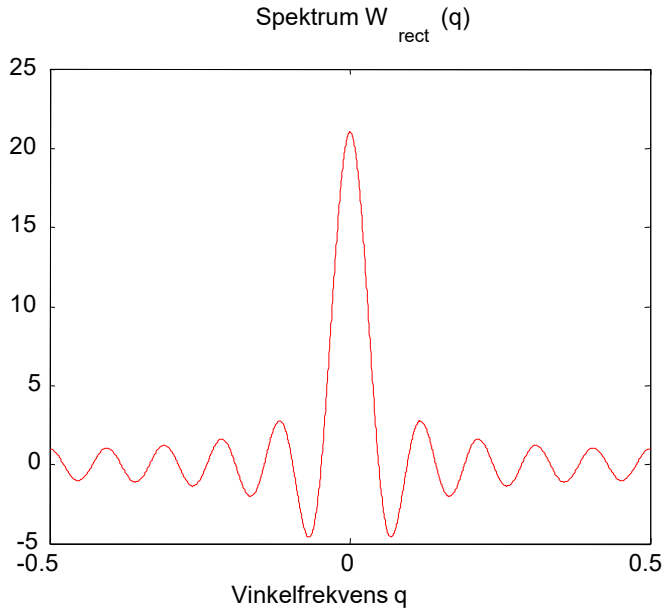
- The wider the window (longer window), the narrower the main lob in frequency domain
  - (It looks more like an impulse)

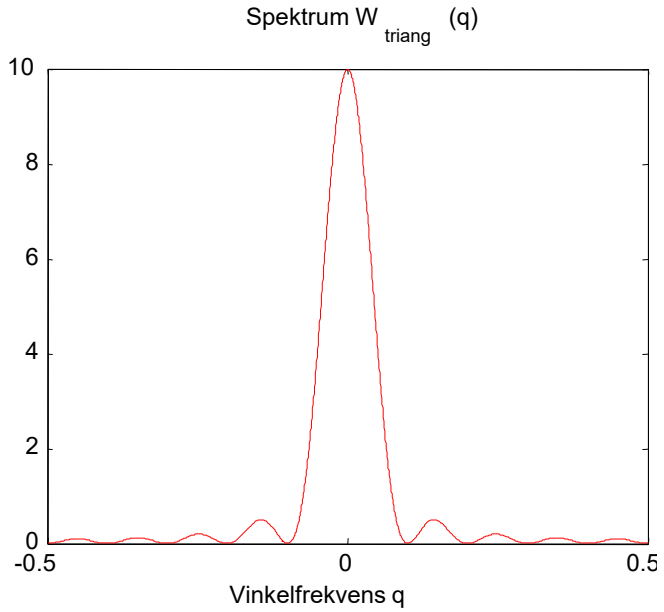


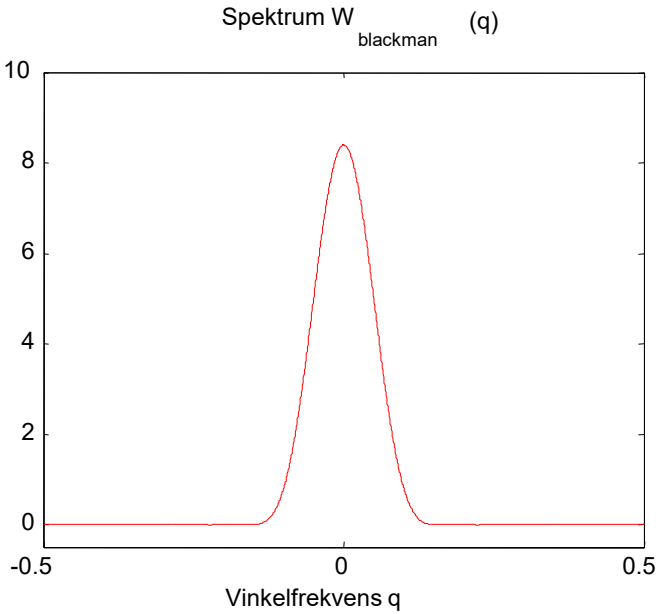


Windowing – Different Window Forms

- Different windows forms imply different damping of sidebands
  - Rectangular, Triangular, Blackman, Hanning, Hamming, etc.





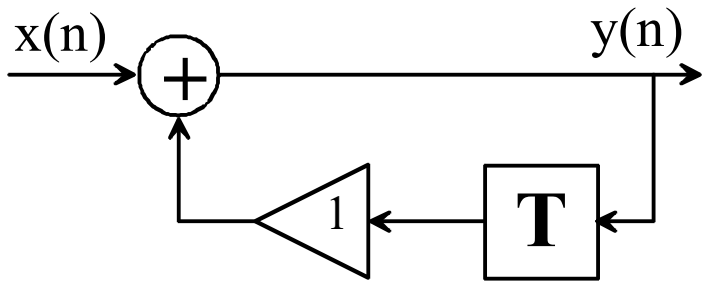


- A narrow main lob implies a poor damping of sidebands

Begränsad delning

Stable Systems

Discrete Time System (Filter) – Example 3



- Feedback (again)
- What is the difference equation for this system?

  $y(n) = x(n) + y(n-1)$

  Assume $x(n) = u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$

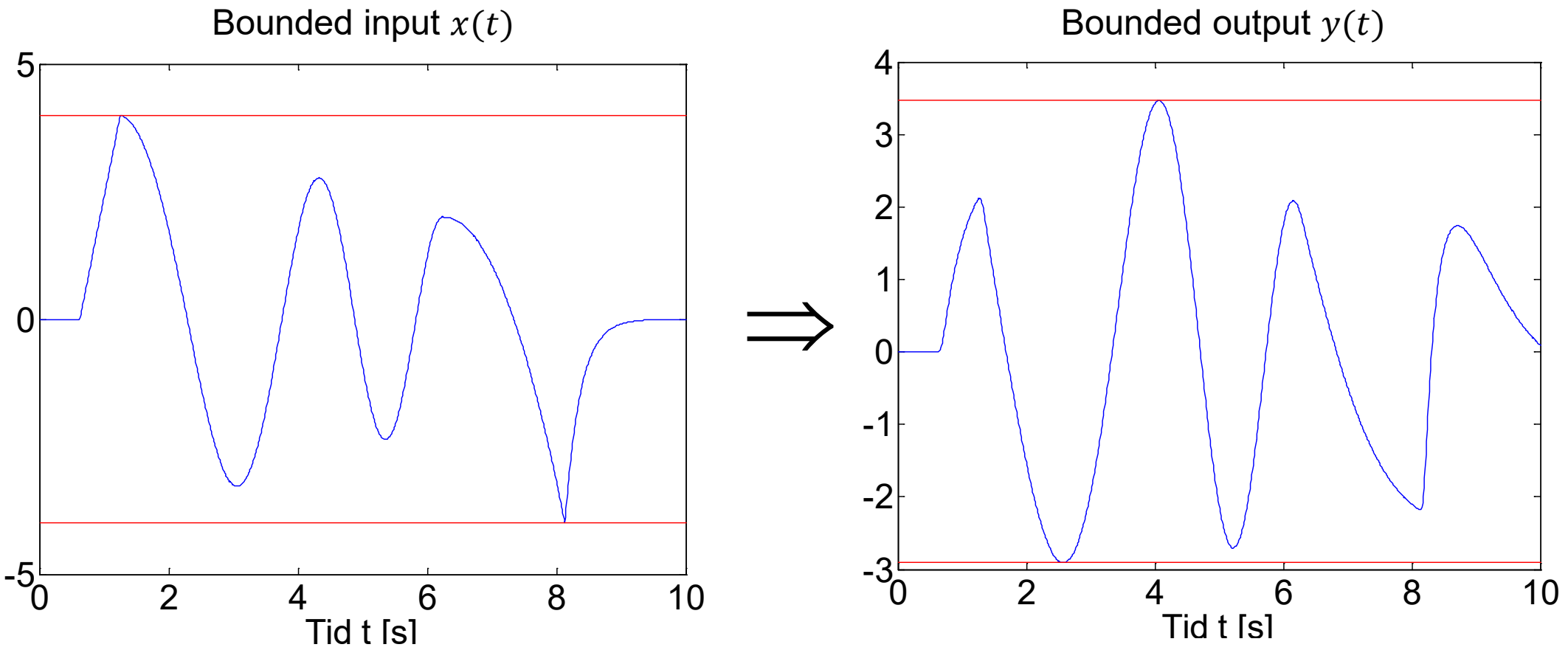
  Note: the input $x(n) \leq C < \infty$ is bounded
- What is the response?

  $y(n) = \sum_{k=0}^{n} (1)^k x(n-k) = \sum_{k=0}^{n} x(n-k) \longrightarrow \infty, \; n \longrightarrow \infty$

  Output is *un*bounded!
- The system is not stable!

- We would like a bounded input to give a bounded output
  - BIBO stability

Bounded Input – Bounded Output (BIBO) stability



BIBO stable system

BIBO stability

- A system is **BIBO stable** if every **bounded input** results in a **bounded output**

$$|x(n)| < K < \infty \Rightarrow$$

$$|y(n)| = \left| \sum_{k=-\infty}^{\infty} h(k) \cdot x(n-k) \right|$$

$$\leq \sum_{k=-\infty}^{\infty} |h(k)| \cdot |x(n-k)| \leq \sum_{k=-\infty}^{\infty} |h(k)| \cdot K < \infty$$

$$\Rightarrow \sum_{k=-\infty}^{\infty} |h(k)| < \infty$$

BIBO Stability for a Discrete-Time System

- Thus, a discrete-time system is BIBO stable if, and only if
  - The impulse response is absolute summable

$$\sum_{k=-\infty}^{\infty} |h(k)| < \infty \qquad \Rightarrow h(n) \to 0, n \to \infty$$
$$\not\Leftarrow h(n) \to 0, n \to \infty$$

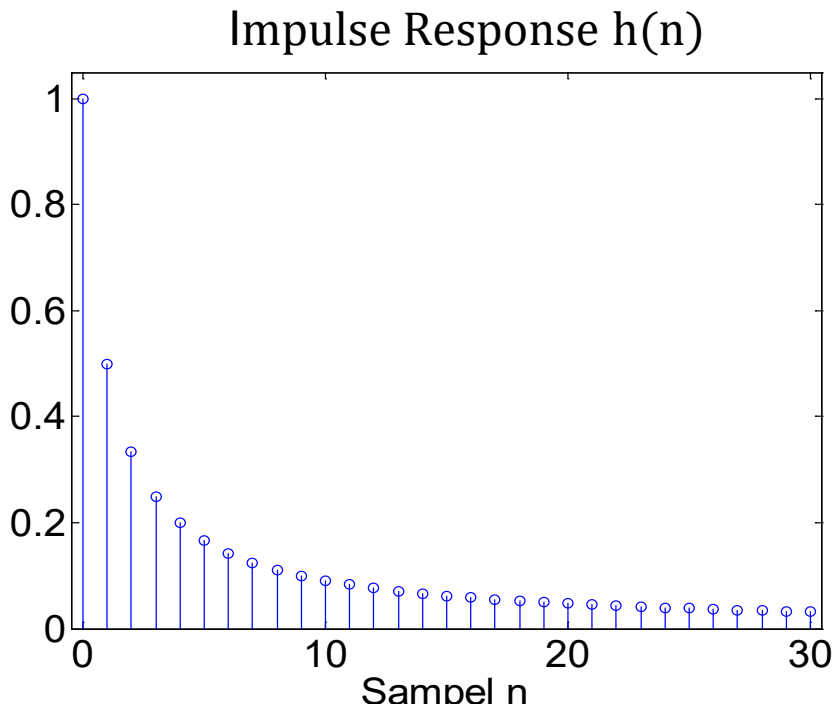
- I.e. if $h(n) \longrightarrow C \neq 0,\ n \longrightarrow \infty$
  - The system is NOT BIBO stable

- BUT if $h(n) \longrightarrow 0,\ n \longrightarrow \infty$
  - The system *may be* BIBO stable (but not surely)

BIBO Stability for a Discrete-Time System – An example

$$h(n) = \frac{1}{n+1}u(n) \to 0, n \to \infty$$

$$\sum_{k=-\infty}^{\infty} |h(k)| = \sum_{k=0}^{\infty} \frac{1}{n+1} \to \infty$$



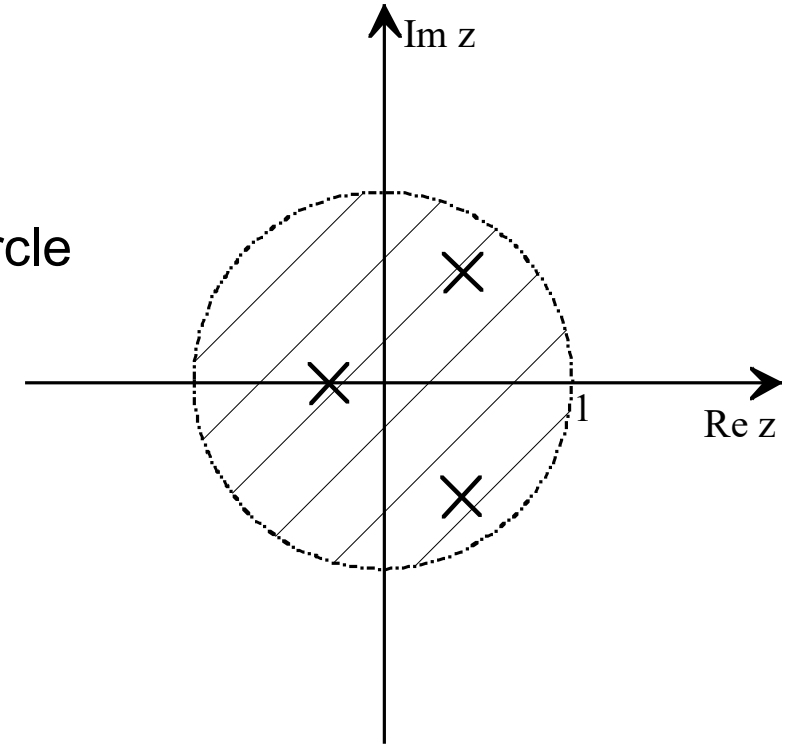
- I.e. the system impulse response $h(n) \longrightarrow 0, \ n \longrightarrow \infty$

  But it is *not* BIBO stable

(Strictly) BIBO Stability for a discrete time system

- A time-discrete LTI system is (strictly) BIBO stable
  if and only if
  all poles of its Transfer function lies within the unit circle

$$\max_i \left| z_{p(i)} \right| < 1$$



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